

Shift-Aware Weighted Conformal Anomaly Detection for Few-Shot Industrial Inspection: Methodology

3 Methodology

This section presents the proposed framework, Shift-Aware Weighted Conformal Anomaly Detection (SW-CAD). The framework converts subspace residual scores computed on frozen self-supervised features into conformal p -values that carry finite-sample, distribution-free guarantees on the false alarm rate, and extends these guarantees to test distributions that differ from the few normal support images through density-ratio weighting. Section 3.1 formalizes the problem. Section 3.2 reviews the subspace residual scoring backbone. Section 3.3 introduces conformal calibration at the patch and image granularities. Section 3.4 presents the shift-aware weighted extension. Section 3.5 states the theoretical guarantees, and Section 3.6 connects the weighting mechanism to post-hoc probability calibration. Section 3.7 analyzes complexity and storage.

3.1 Problem Formulation

Let c denote an inspection category equipped with a support set of k defect-free (normal) images, $\mathcal{D}_k = \{x_1, \dots, x_k\}$, where $k \in \{1, 2, 4, 8\}$ in the few-shot regime considered here. At test time, the system receives an image x drawn from a test distribution and must produce three outputs: (i) a pixel-level (patch-level) anomaly map for localization, (ii) an image-level decision that flags x as anomalous or normal, and (iii) an uncertainty statement attached to each decision. Prior few-shot detectors output raw anomaly scores whose scale is arbitrary and whose implied probabilities are poorly calibrated; thresholds are typically chosen post hoc using labeled anomalies that are unavailable in practice. Our goal is different: for a user-specified level $\alpha \in (0, 1)$, the image-level decision rule should satisfy a finite-sample bound on the false alarm probability,

$$\mathbb{P}(\text{flag } x \mid x \text{ is normal}) \leq \alpha, \quad (1)$$

and, for a user-specified level $q \in (0, 1)$, the set of patches declared anomalous should control the false discovery rate (FDR) at level q . Crucially, these guarantees should be stated (a) without distributional assumptions on the feature or score distributions, (b) under the severe sample constraint imposed by $k \leq 8$ support images, and (c) in a form that degrades gracefully, and measurably, when the test distribution shifts away from the support distribution due to sensor, illumination, or dataset changes.

Two distributions are distinguished throughout. Let P denote the distribution of normal patches (respectively, normal images) in the support domain, from which the calibration data are drawn, and let Q denote the distribution of normal test inputs, which may differ from P under covariate shift. We write $w = dQ/dP$ for the likelihood ratio on the covariate space whenever it exists.

3.2 Frozen Feature Extraction and Subspace Residual Scoring

Following recent training-free detectors, we employ a frozen DINOv2 ViT-S/14 encoder ϕ that maps an image x to a grid of N_p patch tokens $\{f_1(x), \dots, f_{N_p}(x)\}$, $f_j(x) \in \mathbb{R}^d$ with $d = 384$ and $N_p = (H/14) \cdot (W/14)$ at input resolution $H \times W$ (e.g., $N_p = 256$ at 224×224 ; all patch counts below are reported for this operating resolution). The scoring head is deliberately minimal. All patch tokens of the support set are pooled into a matrix $F \in \mathbb{R}^{(kN_p) \times d}$, from which we compute the empirical mean $\mu \in \mathbb{R}^d$ and, via singular value decomposition of the centered features, an orthonormal basis $U_m \in \mathbb{R}^{d \times m}$ spanning the top m principal directions, with $m \in \{64, 128\}$. The anomaly score of a patch feature f is the norm of its residual outside the normal subspace,

$$r(f) = \|(I_d - U_m U_m^\top)(f - \mu)\|_2. \quad (2)$$

The image-level score aggregates the most anomalous evidence within the image,

$$s(x) = \frac{1}{\lceil \rho N_p \rceil} \sum_{j \in \mathcal{T}_\rho(x)} r(f_j(x)), \quad (3)$$

where $\mathcal{T}_\rho(x)$ indexes the top ρ fraction of patch residuals in x and $\rho = 0.01$ throughout. The category model consists only of (μ, U_m) , that is, $d(m+1)$ floating point values, which yields a storage footprint of approximately 0.10 MB for $m = 64$ and 0.20 MB for $m = 128$ in single precision, one to two orders of magnitude below patch memory banks. We emphasize that Eqs. (2) and (3) follow established subspace detectors and are not claimed as contributions; the contribution of this work is the inferential layer constructed on top of them.

3.3 Conformal Calibration at Two Granularities

Conformal inference converts any score function into p -values that are valid in finite samples provided the calibration scores and the test score are exchangeable under the null hypothesis. The few-shot regime makes the choice of calibration unit decisive: with $k \leq 8$ images, image-level calibration alone cannot support small test levels, whereas patch-level calibration offers hundreds to thousands of calibration units per support image at the price of a weaker exchangeability assumption. SW-CAD therefore operates at both granularities and assigns each its proper role.

3.3.1 Patch-Level Conformal p -Values and FDR-Controlled Localization

Out-of-sample calibration scores. A subtle but consequential pitfall must be avoided when forming the calibration set. If the subspace (μ, U_m) is fitted on all support patches and the calibration residuals are computed on those same patches, the calibration scores are systematically deflated relative to the residuals of unseen normal patches, because the subspace has partially absorbed the calibration data. Deflated calibration scores shrink the resulting p -values and inflate the false alarm rate, an anti-conservative failure. For $k \geq 2$ we therefore adopt leave-one-image-out (LOIO) scoring: for each support image x_l , a subspace $\mathcal{S}_{-l} = (\mu_{-l}, U_{m,-l})$ is fitted on $\mathcal{D}_k \setminus \{x_l\}$, and the calibration residuals of the patches of x_l are computed under $\mathcal{S}_{-l}$. For $k = 1$, where LOIO is unavailable, we use a spatially interleaved intra-image split: the subspace is fitted on one half of the patch grid and calibration residuals are computed on the complementary half. The split requires $m < N_p/2$; when this fails (e.g., $m = 128$ at $N_p = 256$), m is reduced accordingly for $k = 1$, and we additionally report the empirical coverage gap of the in-sample alternative.

Conformal p -values. Let $\{r_i\}_{i=1}^{N_{\text{cal}}}$ denote the calibration residuals so obtained, and let $\hat{r}_j = r(f_j(x))$ be the residual of test patch j computed under the full-support subspace at deployment. Scoring test patches under the full-support subspace while calibrating under leave-one-out subspaces introduces a mild asymmetry; because the full-support subspace is fitted on strictly more normal data, the residuals of normal test patches tend to be no larger than their LOIO counterparts, so the mismatch is expected to act in the conservative direction, and the honesty protocol below measures any residual gap. The (unweighted) conformal p -value of patch j is

$$p_j = \frac{1 + \sum_{i=1}^{N_{\text{cal}}} \mathbf{1}\{r_i \geq \hat{r}_j\}}{N_{\text{cal}} + 1}. \quad (4)$$

If $\hat{r}_j$ is exchangeable with the calibration residuals under the null hypothesis that patch j is normal, then p_j is super-uniform, $\mathbb{P}(p_j \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$.

FDR control over the anomaly map. Localization is cast as multiple testing: for a test image, the null hypothesis H_j : “patch j is normal” is tested for every patch simultaneously. We apply the Benjamini–Hochberg (BH) procedure at level q to the collection $\{p_j\}_{j=1}^{N_p}$, declaring anomalous the patches in the rejection set. Conformal p -values computed against a common calibration set are not independent, but they satisfy the positive regression dependence on a subset (PRDS) property, under which BH retains finite-sample FDR control. The output is an anomaly mask with a user-interpretable guarantee: on average, at most a fraction q of the flagged patches are false discoveries.

Exchangeability caveats. Patches within an image exhibit spatial correlation, and patches drawn from the same support image are not independent, so patch-level exchangeability—and the independence of null test scores underlying the PRDS guarantee—holds only approximately. Two mitigations are used. First, calibration may be performed image-conditionally, treating the per-image sets of residuals as the exchangeable blocks and resampling at the block level. Second, and regardless of the variant employed, we report the empirical-versus-nominal coverage gap on held-out normal data for every configuration, so that the practical tightness of the nominal guarantee is measured rather than asserted. This honesty protocol is an integral part of the method.

3.3.2 Image-Level Decisions via Cross-Conformal Prediction

For the image-level decision we retain the image as the exchangeable unit, which is statistically safest but collides with the resolution limit imposed by k . A split-conformal construction would waste support images; we therefore use a cross-conformal (jackknife-style) scheme (Vovk, 2015; Barber et al., 2021) that recycles every support image as a calibration point. For each $l \in \{1, \dots, k\}$, the image score of the held-out support image is computed under the subspace fitted without it, $s_l = s(x_l; \mathcal{S}_{-l})$, and the test image is scored under the same leave-one-out subspaces, $s^{(l)}(x) = s(x; \mathcal{S}_{-l})$. The cross-conformal p -value is

$$p_{\text{img}}(x) = \frac{1 + \sum_{l=1}^k \mathbf{1}\{s_l \geq s^{(l)}(x)\}}{k + 1}, \quad (5)$$

and the image is flagged when $p_{\text{img}}(x) \leq \alpha$. By construction $p_{\text{img}}(x) \geq 1/(k + 1)$, so the achievable operating points are $\alpha \geq 1/(k + 1)$: approximately 0.11 for $k = 8$, 0.20 for $k = 4$, 0.33 for $k = 2$, and 0.50 for $k = 1$. Because cross-conformal p -values are guaranteed only up to a factor of two (Proposition 1), the exactly α -calibrated rule “flag if $p_{\text{img}} \leq \alpha/2$ ” doubles these floors to $\alpha \geq 2/(k + 1)$ —approximately 0.22, 0.40, and 0.67 for $k = 8, 4, 2$, and vacuous for $k = 1$. Rather than obscuring this constraint, we state it as a quantitative finding of the few-shot setting

(Proposition 3) and use it to justify the patch-level pathway of Section 3.3.1, which is the mechanism by which SW-CAD attains fine-grained guarantees despite $k \leq 8$.

3.4 Shift-Aware Weighted Conformal p -Values

The guarantees above presume that test-time normal inputs are exchangeable with the support-domain calibration data. Under covariate shift, corruption, or cross-dataset transfer, this fails and unweighted conformal p -values lose validity, typically uncovering precisely when reliable uncertainty is most needed. SW-CAD restores validity through weighted conformal inference.

Density-ratio estimation on compressed features. Assume the shifted normal distribution Q is absolutely continuous with respect to P with likelihood ratio $w = dQ/dP$ on the covariate space. We estimate w with a lightweight probabilistic domain classifier operating on the PCA-compressed representation $z = U_m^\top(f - \mu) \in \mathbb{R}^m$, which adds no backbone cost and preserves the storage budget. Given calibration covariates $\{z_i\}_{i=1}^{n_0}$ labeled 0 and an unlabeled batch of test covariates $\{z'_u\}_{u=1}^{n_1}$ labeled 1, a regularized logistic model $\hat{g}$ is fitted and the ratio is recovered as

$$\hat{w}(z) = \frac{\hat{g}(z)}{1 - \hat{g}(z)} \cdot \frac{n_0}{n_1}. \quad (6)$$

Because the test batch is unlabeled, it may contain anomalies, so $\hat{g}$ targets the ratio between P and the test *mixture* rather than dQ/dP on normal inputs; we assume the anomalous fraction of the batch is small, and the estimation-error bound of Proposition 2 absorbs the induced discrepancy.

Weighted p -values. Writing z_i for the covariate of calibration unit i and $\hat{z}$ for that of the test unit, the weighted conformal p -value replaces the uniform empirical distribution in Eq. (4) with its importance-weighted counterpart:

$$p^w(\hat{r}) = \frac{\sum_{i=1}^{N_{\text{cal}}} \hat{w}(z_i) \mathbf{1}\{r_i \geq \hat{r}\} + \hat{w}(\hat{z})}{\sum_{i=1}^{N_{\text{cal}}} \hat{w}(z_i) + \hat{w}(\hat{z})}. \quad (7)$$

The same construction applies verbatim at the image level by substituting image scores and image covariates into Eq. (5), where the covariate of an image is taken to be the mean of its compressed patch covariates, $\bar{z}(x) = N_p^{-1} \sum_j U_m^\top(f_j(x) - \mu)$. For multiple testing under shift, BH applied to weighted p -values is no longer protected by PRDS; we therefore adopt the weighted conformal selection (WCS) procedure of Jin and Candès (2025), which restores finite-sample FDR control in the weighted regime.

Effective sample size accounting. Weighting trades validity for statistical resolution. The step size of the weighted p -value is no longer $1/(N_{\text{cal}} + 1)$ but is governed by the concentration of the weights, summarized by the Kish effective sample size

$$n_{\text{eff}} = \frac{(\sum_i \hat{w}(z_i))^2}{\sum_i \hat{w}(z_i)^2} \leq N_{\text{cal}}. \quad (8)$$

Severe shift concentrates the weights, collapses n_{eff} , and inflates the variance of the procedure. SW-CAD reports n_{eff} alongside every shifted evaluation, exposing the coverage-versus-power trade-off as a measured quantity: unweighted inference is sharp but invalid under shift, weighted inference is valid but pays in resolution, and n_{eff} quantifies the price.

3.5 Theoretical Guarantees

We collect the guarantees inherited by the construction, stated deliberately at the level of generality that the few-shot regime supports. Proofs follow from the cited results and are sketched for completeness.

Assumption 1 (Null exchangeability). *Under the null hypothesis, the score of the test unit and the calibration scores $\{r_i\}_{i=1}^N$ are exchangeable random variables.*

Proposition 1 (Finite-sample false alarm control). *Under Assumption 1, the conformal p -value of Eq. (4) satisfies $\mathbb{P}(p \leq \alpha) \leq \alpha$ for every $\alpha \in (0, 1)$ and every sample size; consequently, flagging when $p \leq \alpha$ controls the false alarm probability at level α , without assumptions on the score distribution, and BH at level q applied to the patch-level p -values controls the FDR of the anomaly map at level q , owing to the PRDS property of conformal p -values sharing a calibration set. The cross-conformal p -value of Eq. (5) with exchangeable support images satisfies the factor-two bound $\mathbb{P}(p_{\text{img}} \leq \alpha) \leq 2\alpha$ for every $\alpha \in (0, 1)$; hence flagging when $p_{\text{img}} \leq \alpha/2$ controls the false alarm probability at level α .*

Proof sketch. Exchangeability implies the rank of the test score among the $N + 1$ scores is uniform on $\{1, \dots, N + 1\}$, whence p is super-uniform. FDR control under PRDS follows from Benjamini and Yekutieli (2001), and the PRDS property of conformal p -values is established by Bates et al. (2023). The factor-two bound for the cross-conformal variant follows from the validity analysis of cross-conformal predictors (Vovk, 2015) and of the jackknife+ (Barber et al., 2021); exact validity is not guaranteed in general because no leave-one-out fit includes the test image, and the empirical slack of the factor-two bound is quantified in Section 4. $\square$

Proposition 2 (Validity under covariate shift). *Suppose calibration units are drawn i.i.d. from P , the test unit is drawn from Q with known likelihood ratio $w = dQ/dP$, and the null holds for the test unit. Then the weighted conformal p -value of Eq. (7) satisfies $\mathbb{P}(p^w \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$. When w is replaced by an estimate $\hat{w}$, the coverage inflation is bounded by a total-variation-type discrepancy between the weighted empirical distributions induced by w and $\hat{w}$; in particular, validity degrades continuously in the estimation error of the density ratio. Under the estimated-weight regime, FDR control for the selected patch set is restored by the WCS procedure.*

Proof sketch. The known-weight statement is the weighted exchangeability result of Tibshirani et al. (2019). The perturbation bound follows by comparing the weighted empirical CDFs under w and $\hat{w}$ and applying the stability argument of Lei and Candès (2021); the FDR statement follows from the weighted conformal selection theory of Jin and Candès (2025). $\square$

Proposition 3 (Finite-sample resolution floor). *Any conformal p -value computed from N exchangeable calibration units satisfies $p \geq 1/(N + 1)$ almost surely. Hence at the image level with k support images, no decision rule of the form “flag if $p_{\text{img}} \leq \alpha$ ” can operate at $\alpha < 1/(k + 1)$, and in the weighted regime the attainable resolution is further limited by n_{eff} of Eq. (8) in place of N .*

Proof sketch. Immediate from Eq. (4), whose numerator is at least 1; the weighted statement follows from Eq. (7), whose minimal value is the relative weight of the test unit, which concentrates as the weights degenerate. $\square$

Remark 1. Proposition 3 is not a limitation of the specific estimator but of any distribution-free procedure at this sample size, and it formalizes why patch-level pooling is necessary rather than merely convenient in few-shot industrial inspection. No guarantee is claimed under adversarial perturbation of the test input; adversarial attacks are treated purely as a diagnostic stressor in the experiments, and robust conformal variants are left to future work.

Algorithm 1 SW-CAD: calibration and inference for one category

Require: support set $\mathcal{D}_k$, frozen encoder ϕ , PCA dimension m , levels α (image), q (FDR), unlabeled test batch $\mathcal{B}$

- 1: Fit (μ, U_m) on all support patches; fit LOIO subspaces $\{\mathcal{S}_{-l}\}_{l=1}^k \triangleright$ intra-image split if $k = 1$
 - 2: Compute LOIO calibration residuals $\{r_i\}_{i=1}^{N_{\text{cal}}}$ and image scores $\{s_l\}_{l=1}^k$; sort $\{r_i\}$
 - 3: Fit domain classifier $\hat{g}$ on compressed covariates of calibration data vs. $\mathcal{B}$; set $\hat{w}$ via Eq. (6); record n_{eff} via Eq. (8); fit (a, b) via Eq. (9)
 - 4: **for** each test image $x \in \mathcal{B}$ **do**
 - 5: Compute patch residuals $\{\hat{r}_j\}_{j=1}^{N_p}$ and image score $s(x)$ via Eqs. (2) and (3)
 - 6: Compute weighted patch p -values $\{p_j^w\}$ via Eq. (7); run WCS (BH if unweighted) at level $q \rightarrow$ anomaly mask
 - 7: Compute weighted cross-conformal $p_{\text{img}}^w(x)$ via Eqs. (5) and (7); flag x if $p_{\text{img}}^w(x) \leq \alpha/2$
 - 8: Output (mask, $p_{\text{img}}^w(x)$, $\pi^w(x)$) with π^w from Eq. (9)
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3.6 A Unified Weighting Mechanism for Guarantees and Calibration

The density ratio $\hat{w}$ of Eq. (6) serves a second purpose. Post-hoc probability calibration in this framework maps scores to probabilities through a vector Platt model fitted on calibration data from P ; under shift, the calibration objective is importance-reweighted with the very same $\hat{w}$, yielding the shift-aware calibrated probability

$$\pi^w(x) = \sigma(a^\top \psi(s(x)) + b), \quad (a, b) = \arg \min_{a, b} \sum_i \hat{w}(z_i) \ell(\sigma(a^\top \psi(s_i) + b), y_i), \quad (9)$$

where ψ collects the score features of the vector Platt parameterization, ℓ is the log loss, and y_i are the calibration labels. The two-class calibration set is formed from the normal calibration scores ($y_i = 0$) together with the scores of synthetic pseudo-anomalies obtained by perturbing support patches ($y_i = 1$); without this second class the log-loss fit in Eq. (9) is degenerate. A single estimated object, the density ratio, therefore produces two complementary outputs: a calibrated probability $\pi^w(x)$ for graded decision support, and a valid p -value $p^w(x)$ for guaranteed decisions. This unification is what distinguishes SW-CAD from a mere concatenation of a detector with an off-the-shelf conformal wrapper: the shift-awareness of the probabilistic layer and of the inferential layer is one mechanism, estimated once, audited once (through n_{eff}), and ablated once.

3.7 Algorithm, Complexity, and Storage

Algorithm 1 summarizes deployment for one category. Fitting the subspace costs one thin SVD, $O(kN_p d \min(kN_p, d))$, performed once per category; the LOIO variants multiply this by k , which is negligible for $k \leq 8$. Per test image, feature projection and residual computation cost $O(N_p d m)$, weighted p -values cost $O(N_p \log N_{\text{cal}})$ after sorting the calibration residuals once, and the domain classifier is a logistic model on m -dimensional inputs refitted only when the deployment batch changes. Beyond the subspace parameters (μ, U_m) , the method stores only the sorted calibration residuals (N_{cal} scalars) and their compressed covariates ($N_{\text{cal}} \cdot m$ values), for a total of $(m + 1)(d + N_{\text{cal}})$ floating point values, i.e., $4(m + 1)(d + N_{\text{cal}})$ bytes in single precision. For instance, $k = 8$ support images at $N_p = 256$ patches give $N_{\text{cal}} = 2048$ and a footprint of approximately 0.6 MB per category for $m = 64$ (about 1.3 MB for $m = 128$), preserving a one-to-two order of magnitude advantage over memory-bank detectors.

Scope of Claims

The subspace residual detector on frozen DINOv2 features follows prior art and is not claimed as novel; no claim of state-of-the-art clean AUROC is made; and no robustness guarantee under adversarial attack is asserted. The image-level cross-conformal guarantee holds up to the standard factor of two (Proposition 1). The contributions of this section are (i) the two-granularity conformal construction adapted to the $k \leq 8$ regime, including the out-of-sample scoring protocol, the factor-two accounting of the cross-conformal image rule, and the resolution-floor analysis, (ii) the shift-aware weighted extension with effective-sample-size accounting, and (iii) the unification of distribution-free guarantees and probability calibration through a single density-ratio mechanism.

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